

OUR OPPORTUNITY:

DRIVE THE ENERGY TRANSITION FORWARD

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Q&A



*Arman Koklu is part of the EMEA HDGT **New Units** Project Execution Team as Project **Portfolio Leader for Central & Eastern Europe** providing strategic oversight on projects within the Pole. He is part of the **EMEA Leadership and Execution** team reporting to Mohamad Ali [Project Execution Pole Leader EMEA].*

*Arman joined GE from Siemens in 2013 through the GE Project Management Leadership Program (**PMLP**), taking on various roles in GE Power & Water, Power Services and Renewable Energy businesses based in **New York, US** and **Salzbergen, Germany**. Following the PMLP Arman joined GE Gas Power Systems as a Project Manager in **Europe, Africa & Middle East** region based in Istanbul, gaining extensive project delivery experience in various GE technologies in energy and power generation.*

*Arman has managed multiple projects in North America, Europe, Middle East and Africa putting in total more than 200 Wind, Gas and Steam Turbines into operation in multiple countries such as **US, Canada, Germany, Qatar, Saudi Arabia, Oman and Ghana**. Additionally, Arman was recently named to **Project Management Institute's Future 50** list for his work in helping deliver energy to communities around the world.*

*Arman holds a Bachelor's Degree in **Mechanical Engineering** from Yeditepe University, Istanbul, Turkey and a Master of Science degree in **Power Engineering** from Brandenburgische Technische Universität, Cottbus, Germany. Outside of work, Arman enjoys music festivals and concerts, football both as a fan and a player, traveling and training. Arman will continue to be **based in Baden, Switzerland** where he lives with his wife Cemre.*



- Data Science
- An Innovative Mindset
- Security and Privacy Knowledge
- Legal and Regulatory Compliance Knowledge
- The Ability to Make Data-Driven Decisions
- Collaborative Leadership



IBM's Big Bet on Health / Can Hulu Survive? / Apple's Design Defense

FAST COMPANY

THE SECRETS OF GENERATION FLUX

HOW TO LEAD IN A TIME OF CHAOS

A SPECIAL REPORT BY ROBERT SAFIAN

Illustration: David LaChapelle
Left: Photo: David LaChapelle
Right: Photo: David LaChapelle
Photo: David LaChapelle
Photo: David LaChapelle



A Post-Pixar Hit / Can a Printer Replace a Factory? / The Doctor in Your iPhone

FAST COMPANY

MODERN BUSINESS
IS PURE CHAOS.
BUT THOSE WHO ADAPT
WILL SUCCEED.

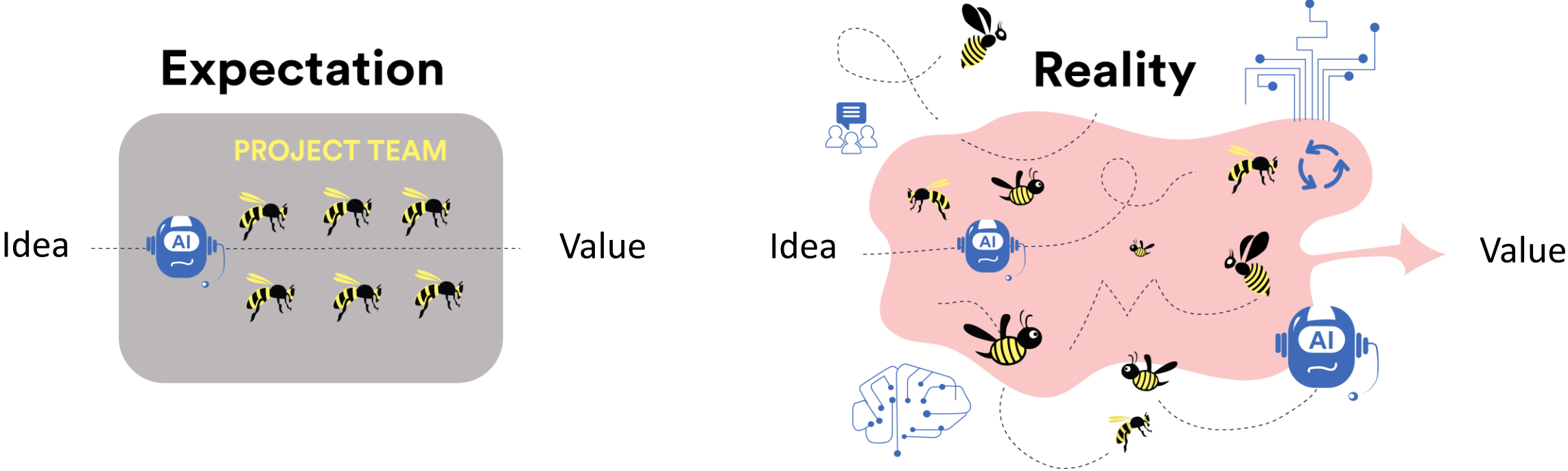
THE SECRETS OF GENERATION FLUX

A SPECIAL REPORT BY ROBERT SAFIAN

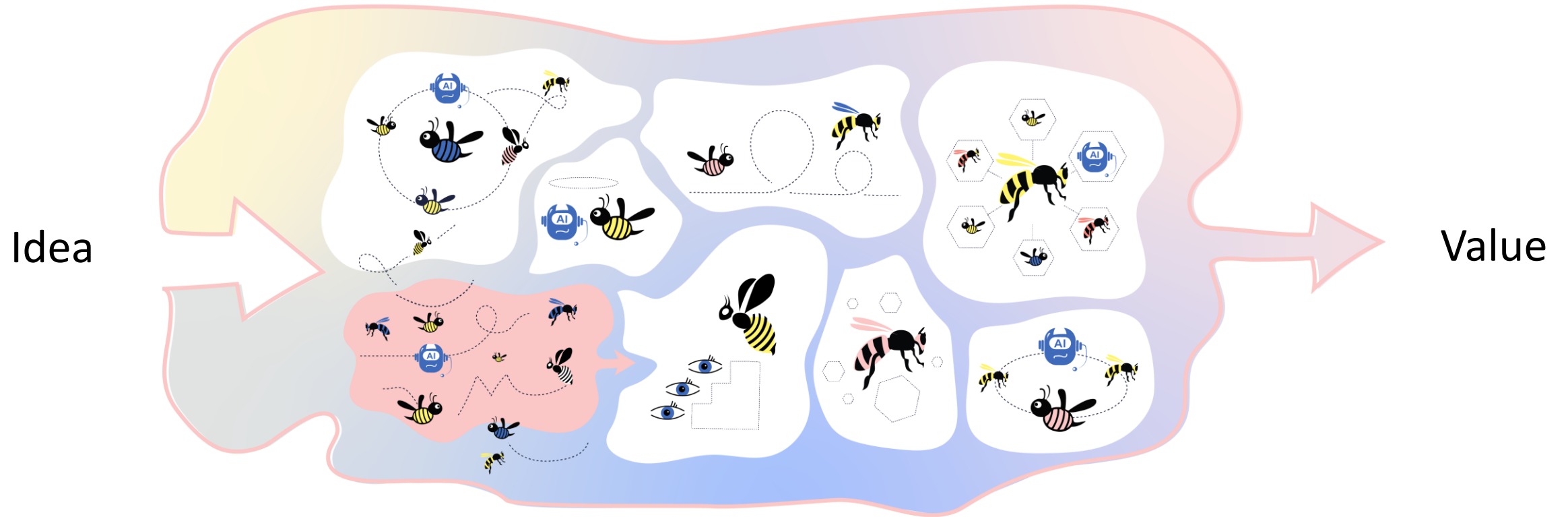


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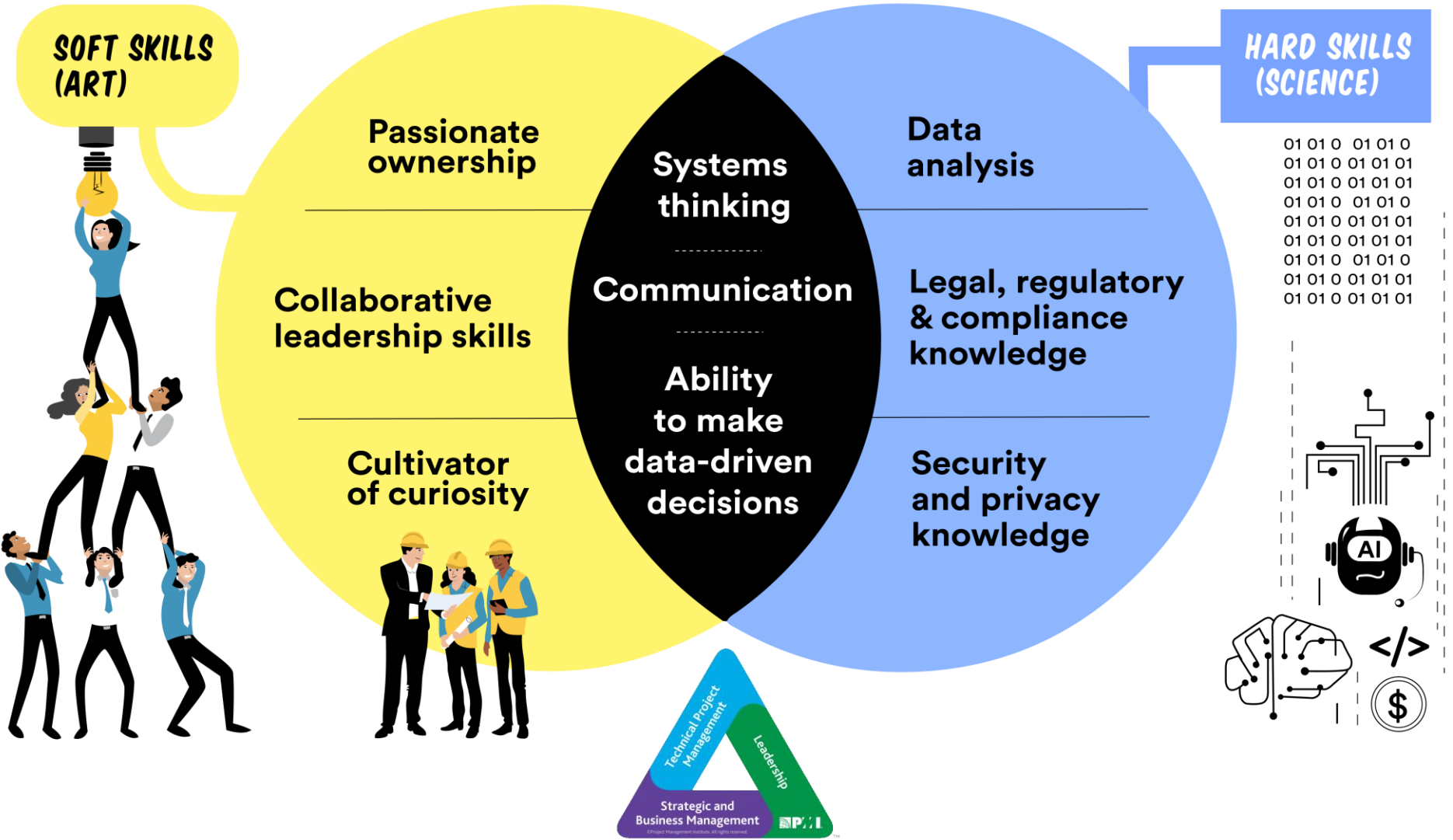
EXPECTATION VS. REALITY



ADAPTIVE SYSTEM



People



WE ARE



GE VERNOVA

For the new era of energy...a new company with full focus on the energy transition



~ **75K**

Global employees

~ **100**

Countries

POWER

Gas Power, Hydro Power, Nuclear, Steam Power

WIND

Onshore Wind, Offshore Wind, Wind Turbine Blades

ELECTRIFICATION

Electrification Software, Grid Solutions, Power Conversion & Storage Solutions

ACCELERATORS

Advanced Research, Consulting Services, Financial Services

Our purpose: The Energy to Change the World

GE Vernova, the 132-year-old energy pioneer. Our unwavering commitment to powering the planet, now fully focused on the energy transition. No company has the unique combination to pioneer what's next – just as we have before. Together, with our customers and partners, we will chart the course to a better future.

GE VERNOVA VIDEO

75,000
employees

140+
countries

132
years of innovation

POWER

Electrifying

Delivering firm, dispatchable power

800 GW

world's largest
installed base of
gas turbines

25 %

of global hydropower
generating capacity

60 GW

service & fuel
for the nuclear
installed base

400 GW¹

servicing critical
baseload steam
power

Solving Energy Trilemma

Energy Security

Affordability

Sustainability

Decarbonizing

Breakthrough technologies

- Capable to operate with up-to **100% hydrogen** in certain gas turbines
- Driving carbon reduction by increased efficiency, life extension & upgrades
- **Carbon Capture** & Storage (CCS)
- Development of Nuclear **Small Modular Reactor** (SMR)

GAS POWER



- Heavy Duty GT
- Aeroderivative GT
- Steam Turbines, Generators

HYDRO POWER



- Hydro Turbines/Generators
- Pumped Storage

NUCLEAR POWER



- Boiler Water Reactors and Small Modular Reactors (SMR): fuels and services
- SMR

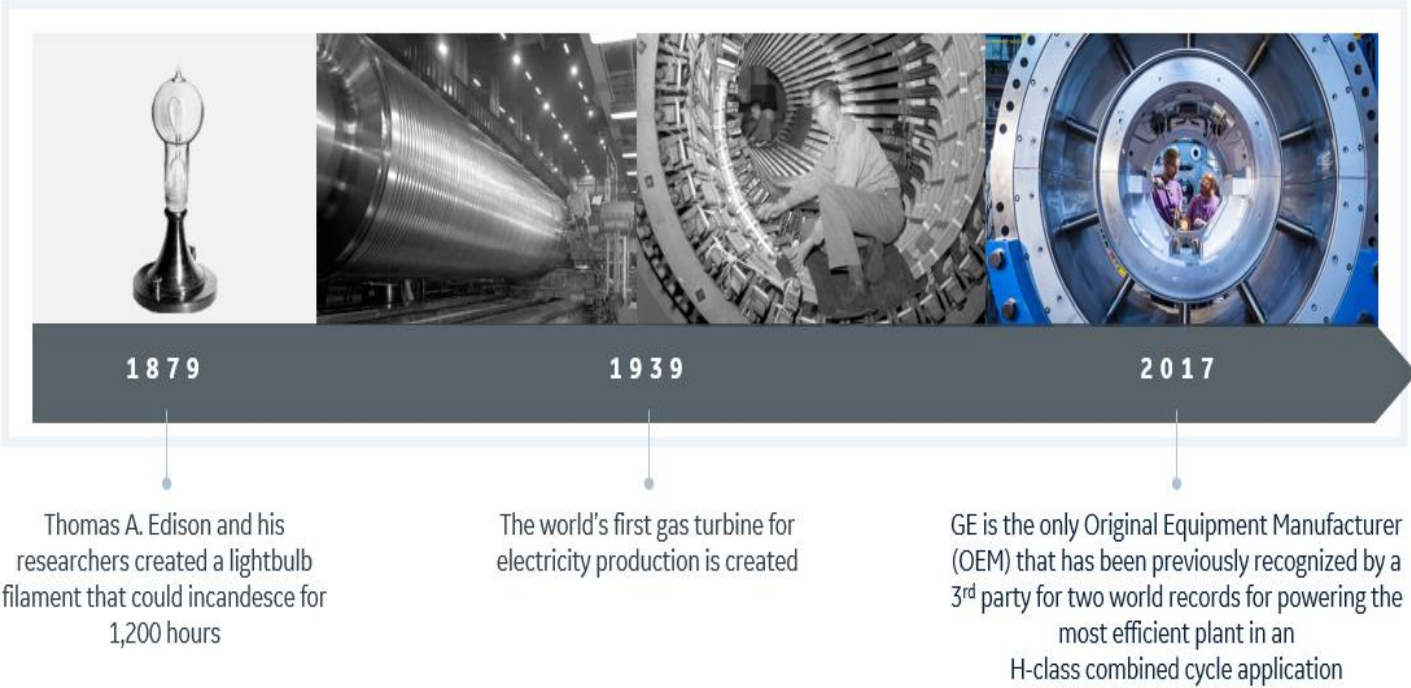
STEAM POWER



- US Nuclear
- Global Coal Services
- Steam, Generators, Boilers

1. excludes portion of the Steam business's nuclear activities sold to Electricité de France S.A. ("EDF"); Decarbonization in this presentation is intended to mean the reduction of carbon emissions on a kilogram per megawatt hour basis

Our history...it all started with the lightbulb



Our mission has evolved over time

-  The Guarantee of Excellence on Goods Electrical
-  Largest Electrical Manufacturer in the World
-  Live Better Electrically
-  Progress Is Our Most Important Product
-  We Bring Good Things to Life
-  Imagination at work
- **Building a world that works**
- **GE VERNOVA**
- The energy to change the world

The World Needs More Power that's Sustainable, Affordable, Reliable & Secure

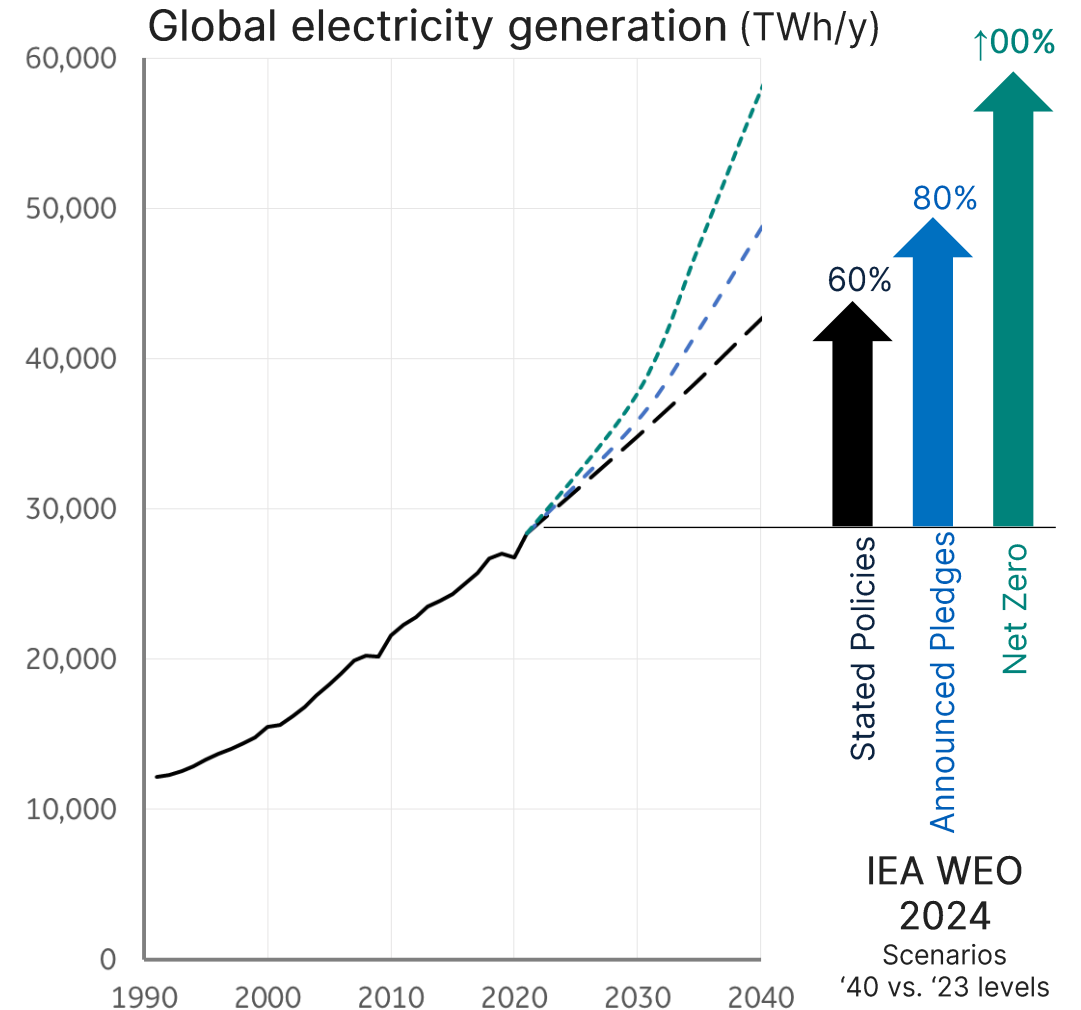
Electricity demand growth continues

- >80% growth projected over the next 2 decades
- ~750 M people lack access to reliable electricity
- Key enabler of economic growth, health & prosperity
- Electrification of transportation, industry and heat to decarbonize non-power sectors

GE Vernova's technology base helps
generate ~25% of the world's electricity



- ▶ we must help electrify the world
- ▶ while decarbonizing it with lower-carbon technology
- ▶ that is sustainable, affordable, reliable/secure



Challenges + opportunities

More Sustainable Energy Investment Super-Cycle

- Underway
- Binding targets driving investment
- ESG expectations
- Government mandates

Evolving Generation Mix

- Wind and solar PV are expected to provide 30% of global electricity by 2030
- Continued reliance on gas to ensure reliability

Demand Growth*

- ~775 million people without reliable electricity today
- Expected 50% increase by 2050
- The energy transition will require cumulative investment of \$150 trillion in transition-related assets from 2023 up to 2050

Energy Security & Stability

- Geopolitical tensions across the globe
- Smart grid and digital to manage integration, decentralization
- Supply chains risks

Electrification

- Global emissions reduction targets
- Increased consumption across more uses
- Storage & infrastructure challenges
- Drives economic growth

Abatement

- Maintaining efficiency
- Widespread hydrogen adoption + profitability timing uncertain
- Carbon capture taking off slowly
- Hard-to-abate industries decarbonization

\$435B^{a)}

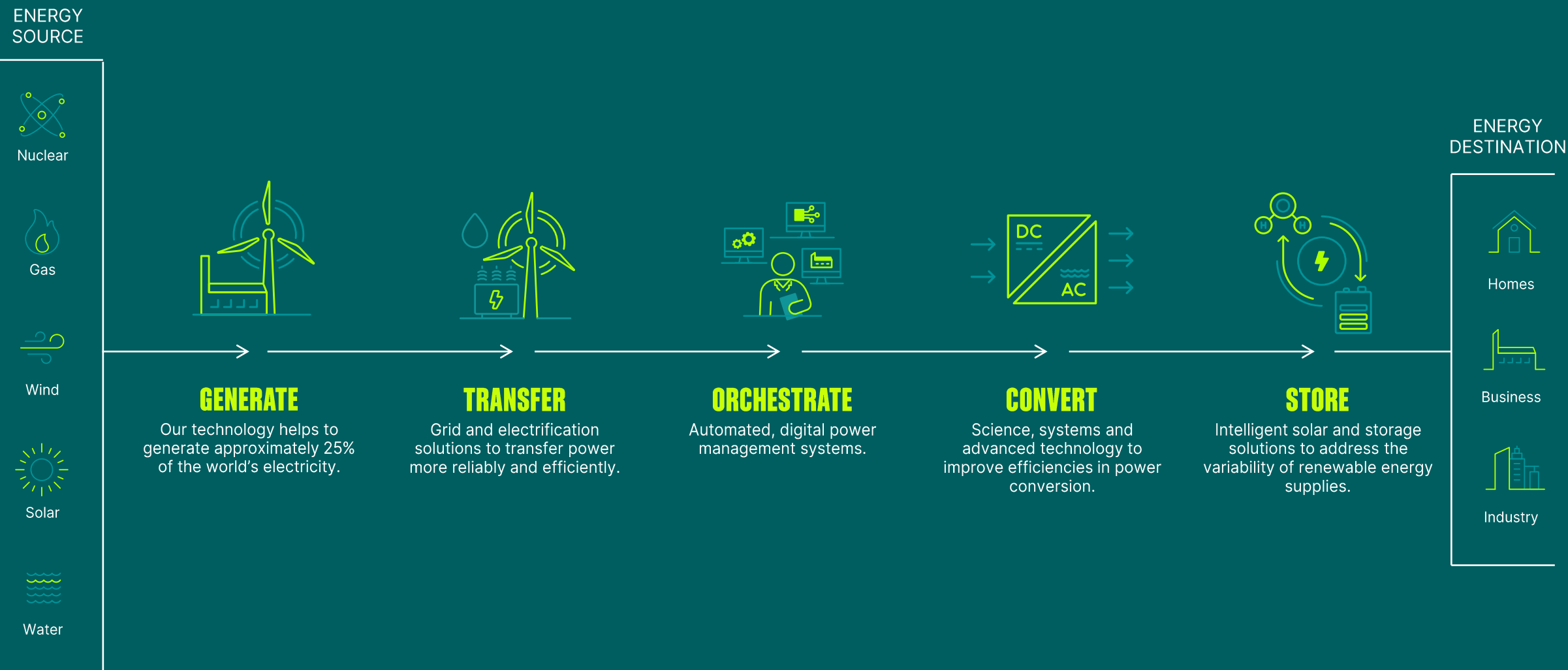
ADDRESSABLE
GLOBAL DEMAND
THROUGH 2050

(a) GE Vernova estimate of the served available segment, including capital expenditures (CAPEX) and services.

OUR MISSION:

**ELECTRIFY THE WORLD
WHILE DECARBONIZING IT**

Uniquely Positioned to Help Lead the Energy Transition



Strength + reach



~ 25%

World's electricity generated with the help of our technology



1st

Small Modular Reactor commercial contract signed in North America



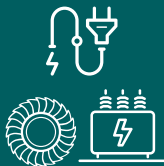
~ \$35B

2024 revenue



~ 7K

Gas turbines installed... world's largest fleet



1st

Enhanced Electric Gas Turbine (EGT) Aero + Storage + Hybrid Control



~ \$44B

Orders



~ 57K

Wind turbines installed in >50 countries... #1 Onshore Wind in US^a



+95%

of power transmission utilities rely on our Electrification Systems components



~ \$1B

Investment in Research & Development



+2,200GW

Global Installed Base



90%

Of companies across the sectors use our industrial software



\$119B

Backlog*

Driving the energy transition



POWER

GAS POWER

- Gas turbines for utilities, power producers & industries, from mobile units to plants.
- Full maintenance & lifecycle services.

HYDRO POWER

- Efficiency upgrades increase lifecycle and TWh of renewable hydro
- Pumped storage projects enable grid stability

NUCLEAR POWER

- Nuclear solutions for boiling water reactors, fuel, services & small modular reactor development with Hitachi joint ventures.

STEAM POWER

- Upgrade/extend life of nuclear plants and maintain a largely renewable industrial fleet in the Americas
- Support global customers with best-in-class services including end of life as they transition away from coal

WIND

WIND BLADES

- Blade technology vital to higher capacity factors
- Two-piece blades reduce transport logistics (and associated indirect emissions)

ONSHORE WIND

- Provides onshore wind turbines & services, optimizing performance with digital tech for cost, capacity & efficiency improvements.

OFFSHORE WIND

- Provides offshore wind tech & farm development, featuring the Haliade-X 220m turbine.

ELECTRIFICATION

ELECTRIFICATION SOFTWARE

- Orchestrates the complexity of the sustainable energy grid while driving reliability and resilience
- Accelerates decarbonization across the energy lifecycle
- Optimizes the measurement and reporting of progress to net-zero goals

ELECTRIFICATION SYSTEMS - GRID SOLUTIONS

- Enables necessary grid modernization, expansion, stability and flexibility to renewables growth in the power system

ELECTRIFICATION SYSTEMS - POWER CONVERSION & STORAGE

- Enables decarbonization of mission-critical industrial applications.
- Batteries and hybrids extend dispatch of renewables enabling reduction of fossil generation for short-durations.

ACCELERATORS

ADVANCED RESEARCH

- A hub for innovation, collaborating with top energy thought leaders to develop and demonstrate breakthrough technologies that enable the deployment of reliable, affordable, and sustainable energy solutions at scale

CONSULTING SERVICES

- Policy, planning and investment decision expertise to help with holistic decarbonization planning.

FINANCIAL SERVICES

- Financing and securing tax equity are key enablers for zero-carbon projects
- Project development and carbon monetization schemes vital to early carbon capture demonstration pilots

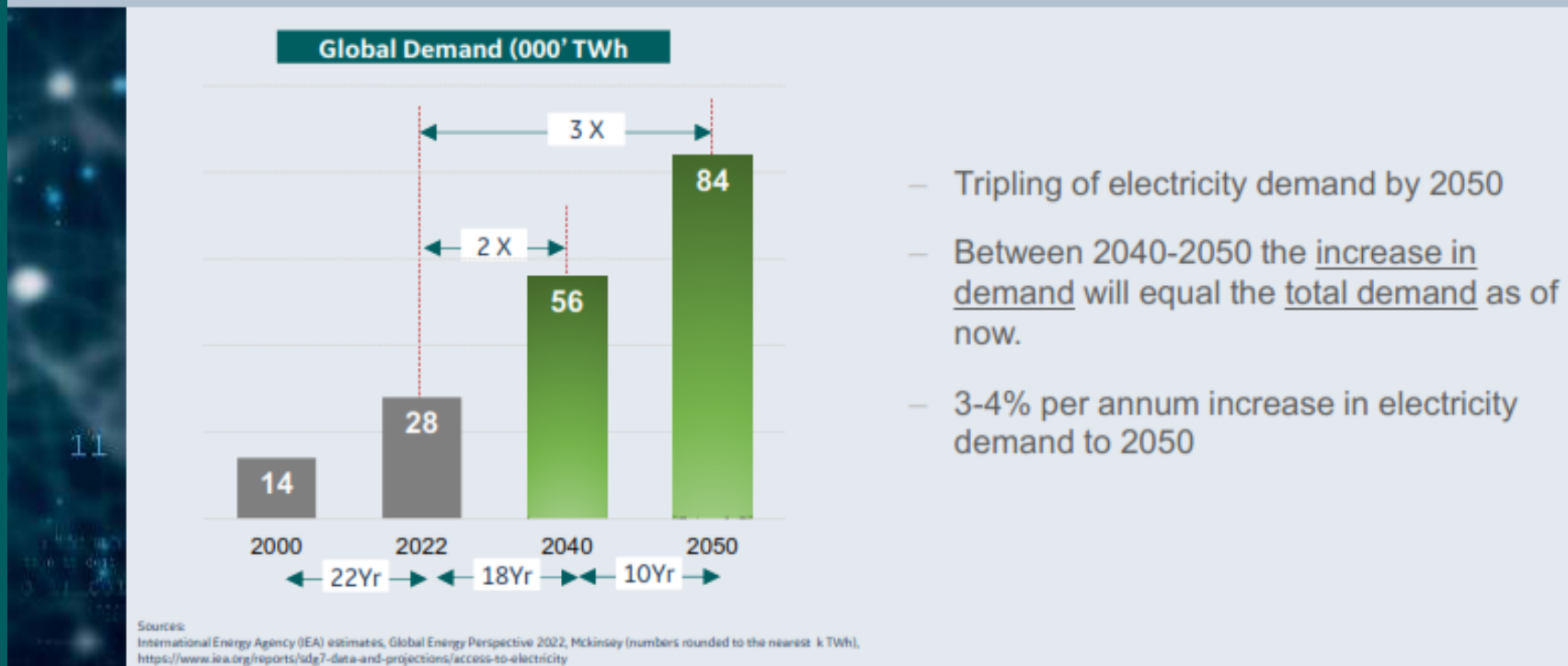
OUR PURPOSE:

**THE ENERGY TO
CHANGE THE WORLD**

Key topics to share:

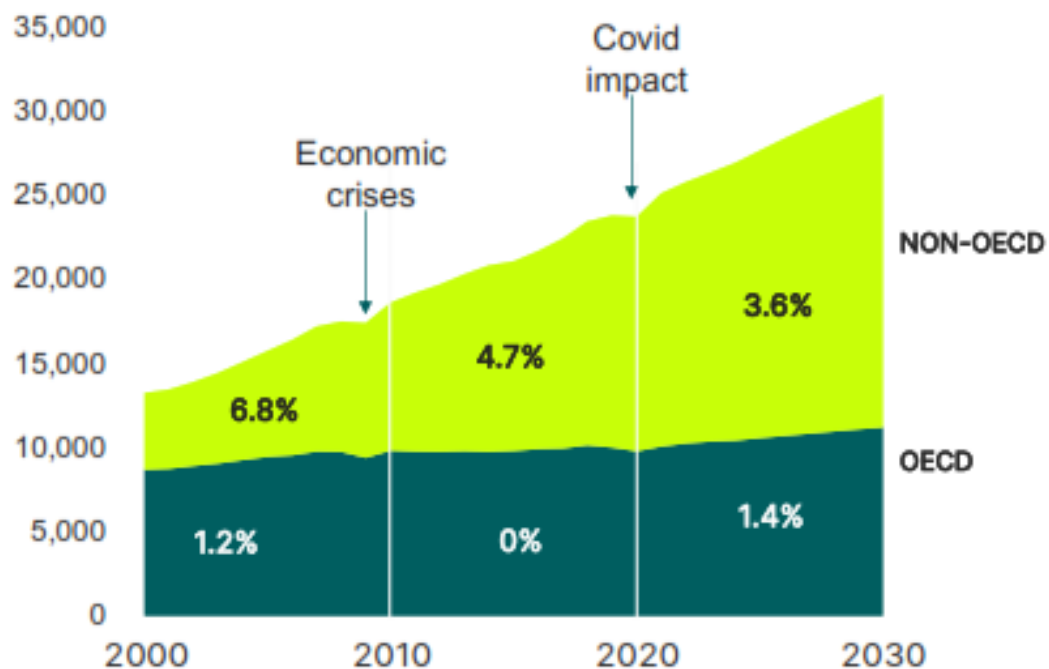
- Electricity demand trends
- Decarbonisation & Net Zero Targets
- Development of Generation & Grid
- System Impacts and Challenges
- Summary & Conclusion

Demand for electricity will increase Significantly



Factors affecting electricity demand

Global electricity consumption (TWh/y)



FACTORS TENDING TO INCREASE CONSUMPTION

- Energy access... more people connected
- Shift towards industrial economies
- GDP growth & expanding middle class
- Data centers and computing power
- Electrification of transport/heat

FACTORS TENDING TO DECREASE CONSUMPTION

- Energy efficiency... LED lights, appliances, buildings
- Shifts from industrial towards services economies
- GDP contraction, financial crises

Source: Global Power Outlook 2022

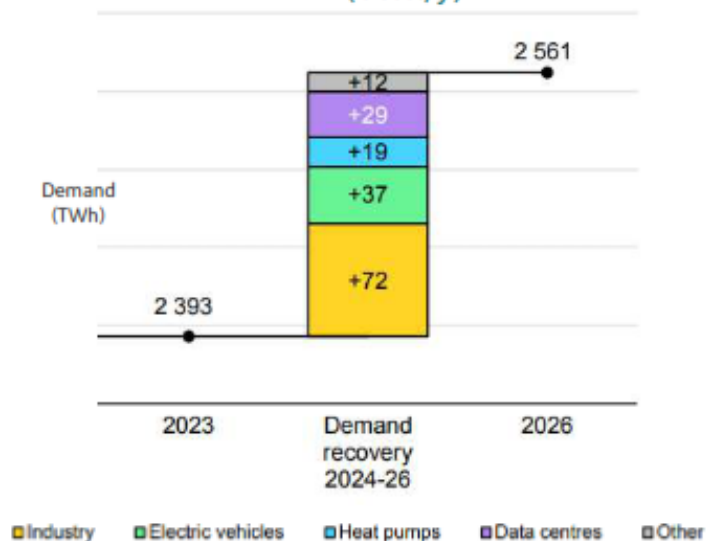
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Drivers of electricity demand in the near-term



Electricity consumption in European Union (TWh/y)



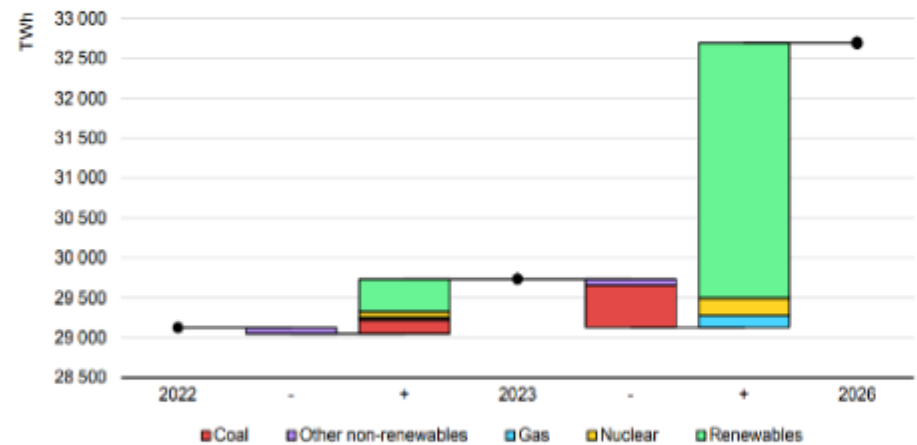
- 55% new demand, of which:
 - 20% is data centres
 - 35% electrification of transport & heat
- 40% is industrial demand recovery

Sources: International Energy Agency (IEA), Electricity 2024 - Analysis and forecasts to 2026

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Changes in global electricity generation, 2022-2026



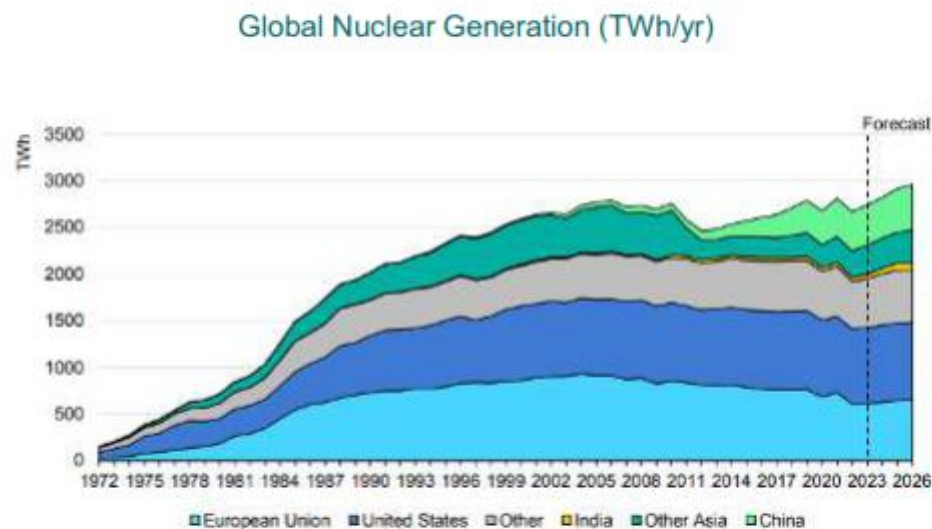
Renewables will overtake coal as the largest source of global electricity supply in 2025

China commissioned as much solar PV capacity in 2023 as the entire world did in 2022

Renewable generation expected to grow by around 20% in 2024

Southeast Asia, amid significant demand growth, coal-fired generation is set to increase ~ 4% /year out to 2026.

Resurgence of nuclear generation – 1/2



Source: IEA analysis based on data from World Nuclear Association.

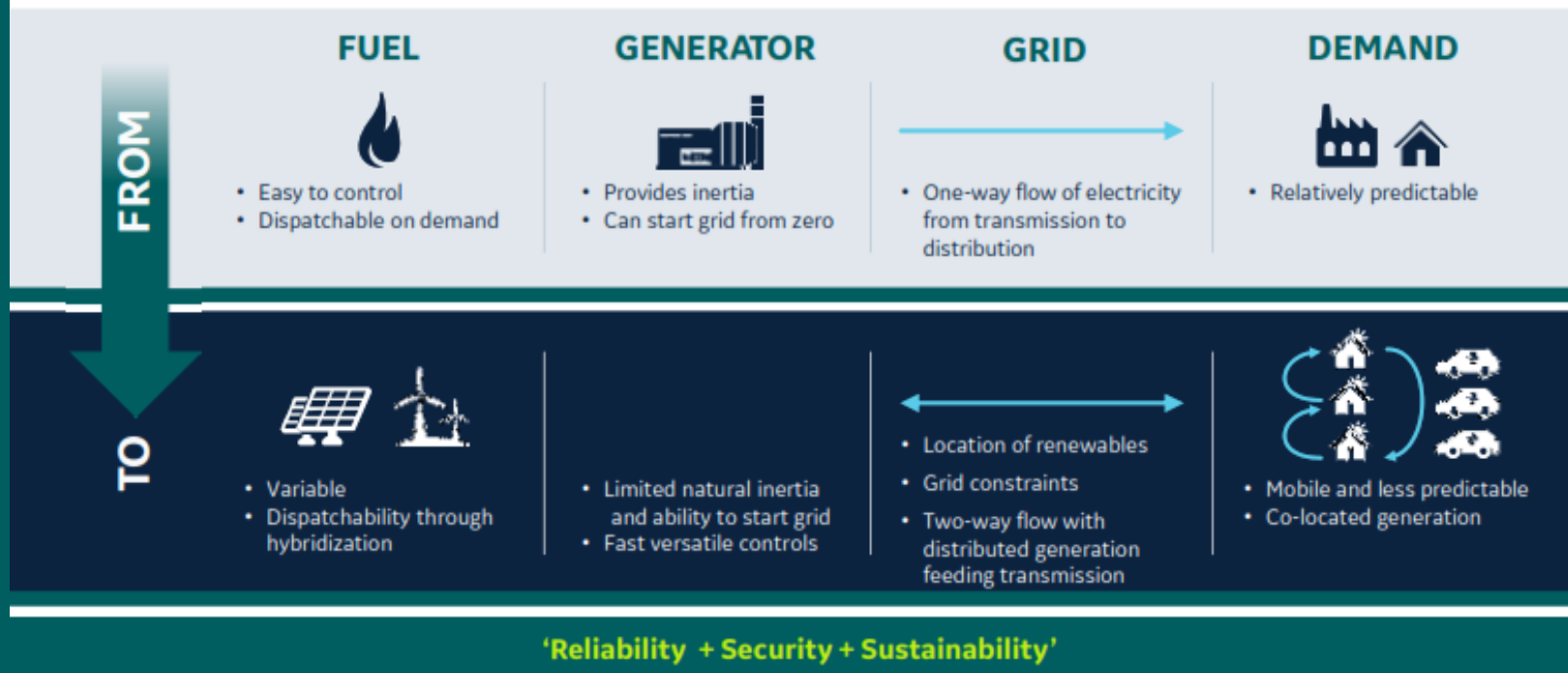
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Global Nuclear generation will be almost 10% higher in 2026 compared to 2023 – new plants starting commercial operation in various regions, as well as French nuclear recovery and expected restarts in Japan.

Nuclear energy more than doubling by 2050 – IEA's updated Net Zero Roadmap.

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Grids - the backbone for secure energy transition



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Summary



- ✓ Electricity demand will increase 3X by 2050
- ✓ Generation capacity will evolve to be predominantly renewables & other clean technologies
- ✓ Maintaining system reliability will need dispatchable capacity (gas as a transition fuel) and storage as sources of flexibility, as well as more digital & active control of the system
- ✓ Large investments in the grid will be required to deliver Decarbonisation/NZ targets – the equivalent of the size of current global grid size needs to be build

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*“Fortune Favors the Prepared Mind”
(Louis Pasteur)*

Thank You

Wir sind

Nosotros somos

نحن

МЫ

CHÚNG TÔI LÀ

Nous sommes

Είμαστε

我们是

TÁIMID

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GE VERNOVA

WE ARE

Ние

私たちはです

Сме

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MES ESAME

Mi vagyunk

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My jsme

اسان آهيون

Biz

Amra achi

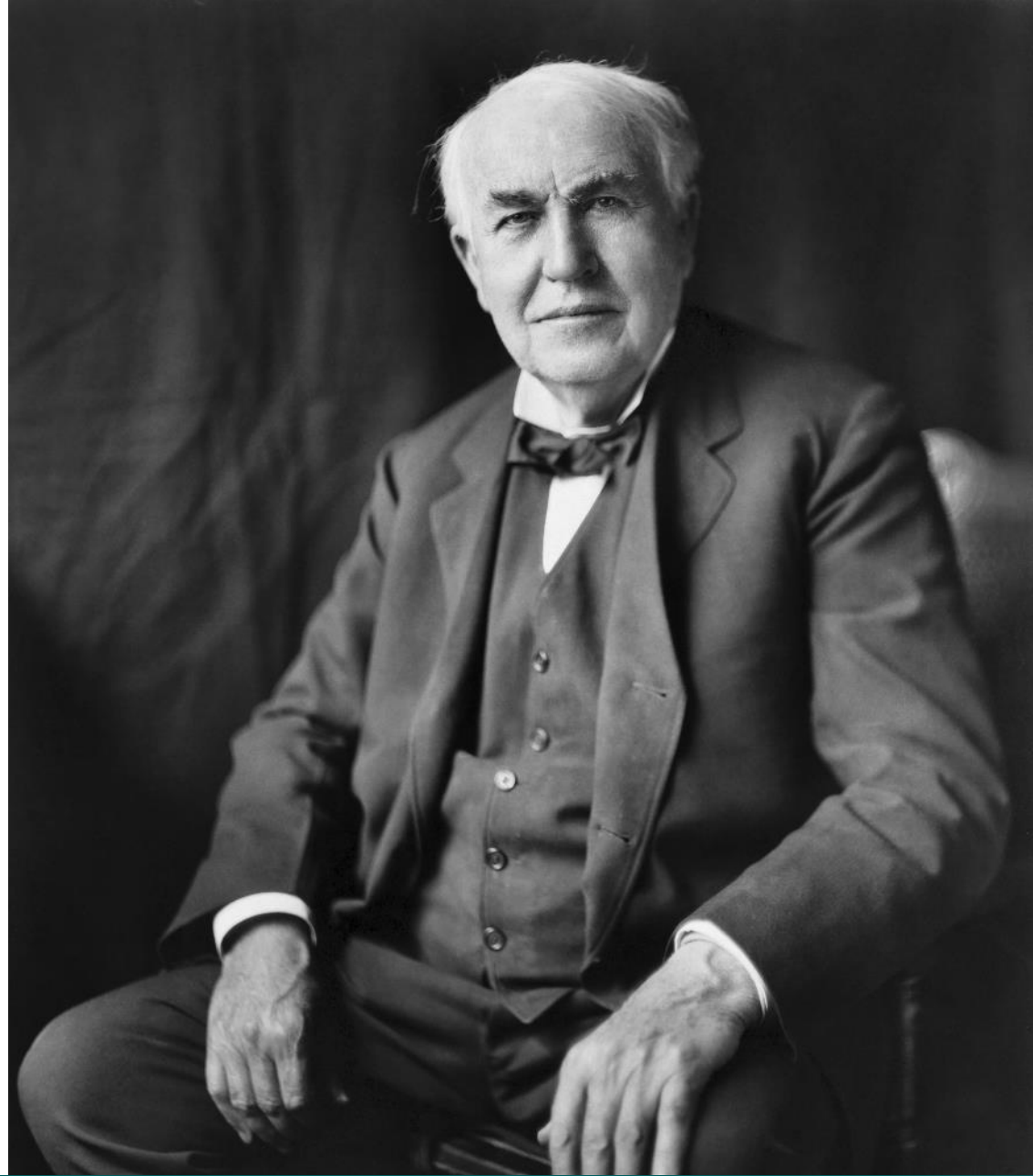
Hum hai



GE VERNOVA

“When you have
exhausted all
possibilities,
remember this...
YOU HAVEN'T.”

Thomas A. Edison
(1847–1931)



WELCOME TO THE ENERGY OF CHANGE



The relentless optimism that creates breakthrough innovation is what fuels GE Vernova to deliver the energy technologies the world needs.

[Explore now](#)